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Leveraging remote sensing and crowd-sourced biodiversity data for enhanced plant functional trait mapping

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High-resolution maps of plant functional traits are crucial for understanding terrestrial ecosystem processes; however, their integration into ecosystem models has been hindered by uncertainties and a lack of spatially detailed data. Here we combine optical remote sensing, global crowd-sourced biodiversity records and plant trait databases to map community trait distributions worldwide at 1-km resolution, estimating community-weighted means (CWMs) and higher-order moments (standard deviation, skewness, and kurtosis) for specific leaf area (SLA), leaf nitrogen (LNC) and leaf phosphorus (LPC) concentrations. Benchmarking against sPlotOpen plot-level CWMs shows low explained variance (R^2 = 0.10–0.27 across traits), indicating limited plot-scale predictive skill under current limited open global benchmarks and scale mismatches. Agreement increases when using a canopy-weighted comparator (TWM; R^2 = 0.22–0.38; relative RMSE ≈ 12–18%), consistent with the top-of-canopy sensitivity of optical sensors. By providing spatially explicit trait distributions and their higher-order moments, our findings deliver improved detail for understanding biodiversity patterns and ecosystem functioning and provide landscape-scale insights into trait-mediated coexistence. This work enhances ecological modeling and offers a foundation for assessing the impacts of global environmental changes, advancing our understanding of plant functional diversity's role in ecosystem resilience and sustainability.

Earth system models (ESMs) require representations of coupled climate-biosphere processes. In their land components, vegetation functional traits encode plant controls on carbon, water, and energy fluxes. By quantifying observable aspects of morphology, physiology, and phenology, trait information helps constrain processes such as nutrient cycling and carbon sequestration[1–5]. Despite their importance, ESMs traditionally rely on plant functional types (PFTs), which aggregate diverse species traits into broad categories. Traditional ESMs also parameterize traits using mean values to represent trait distributions, which can misrepresent ecosystem responses due to nonlinearities in vegetation processes[6]. These assumptions risk oversimplifying the variability in vegetation composition and structure, reducing the

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accuracy of vegetation response modeling and restricting our ability to understand and predict ecosystem feedbacks on climate[7–9].

Addressing the previous problems requires a more nuanced representation of trait variability in ESMs[10–12]. Incorporating higher-order moments beyond averages (e.g., variance or skewness) to describe trait distributions in ESMs and dynamic global vegetation models (DGVMs) improves simulations of vegetation carbon stocks and fluxes across environmental gradients[13–16]. However, the trait data needed for such advancements remain spatially and ecologically incomplete due to uneven observation coverage, bias toward accessible regions, and underrepresentation of certain ecosystems[17–19].

Generating global trait maps involves scaling up individual-level measurements to ecosystem-level community-weighted means (CWMs) and creating geospatially resolved, gap-free maps using regression-based methods[20]. Researchers have developed various approaches to generate static global estimates of traits such as specific leaf area (SLA), leaf nitrogen (LNC), and phosphorus concentrations (LPC) using limited in-situ data[21–28]. These diverse methods have led to divergent outcomes that complicate direct comparison and evaluation[20].

Recent studies have leveraged extensive biodiversity crowd-sourced occurrence databases to calculate CWMs of critical plant traits at a broad scale[26,28]. However, deriving species abundance from presence data remains a critical challenge, often limited by significant sampling biases due to the opportunistic nature of data collection in crowd-sourced initiatives[29–31].

This research addresses two critical limitations in global ESM trait parameterization: (1) the oversimplification of vegetation diversity via PFTs, and (2) the lack of detailed pixel-level, high-resolution functional trait distributions beyond mean values. Supported by extensive sampling and relevance to land-model fluxes, we estimate SLA, LNC, and LPC and compute community means and higher-order moments (1 km grid) to improve understanding of plant performance and ecosystem functioning[4,32,33]. Using species-occurrence data from the Global Biodiversity Information Facility (GBIF)[34] and trait data from the TRY Plant Trait Database[2,18], we estimate species abundances and describe trait variation. GBIF provides over one billion geolocated biodiversity records, while TRY includes trait data for more than 250,000 species. To correct for GBIF sampling biases, we integrate remote sensing data with multiresolution aggregation techniques, thereby improving both spatial resolution and representativeness.

sPlotOpen is a large, open-access compilation of vegetation plots. Although its geographic and biome coverage is uneven, it remains the most readily usable global resource for abundance-based, plot-level trait estimates, and we use it as a benchmark for cross-dataset comparisons[19]. Against this benchmark, our spatially explicit maps of trait distributions can support macroecological analyses and provide priors and benchmarking targets for trait-based model parameterization, while reflecting canopy-weighted, dominant-vegetation signals captured by optical remote sensing.

Results

Global trait maps and associated distribution metrics

Figure 1a shows maps of CWMs for three key leaf traits: SLA, LNC, and LPC. The weighted mean of a normally distributed trait represents the most common value in a given environment, often referred to in ecology as functional dominance. Consistent with prior work, we analyze trait values on the natural-log scale to stabilize variance[2,22,28,33]; for interpretability, estimates and summaries are back-transformed to their original units where appropriate. In particular, mean trait maps are shown on the original scale to facilitate the interpretation of spatial patterns. Interpretation of regional patterns should be guided by the quantitative benchmarking against sPlotOpen (see the Comparison with the sPlotOpen database section in the Results), where plot-level agreement is constrained by scale mismatch and the uneven, incomplete coverage of this global open benchmark. The SLA global distribution, defined as the ratio of leaf area to leaf dry mass, aligns with existing research and recent publications[20]. As shown in Fig. 1a, SLA exhibits broad-scale spatial structure, with higher values in tropical and subtropical regions and lower values in high latitudes. SLA typically ranges from 15 to 25 $mm^2\ mg^{-1}$, with moderate to high values (~15$mm^2\ mg^{-1}$) observed in tropical rainforests and other warm, humid environments. In contrast, cold or dry regions such as deserts, boreal forests, and tundra exhibit low SLA values (<7 $mm^2\ mg^{-1}$).

LNC estimates, defined as the ratio of leaf nitrogen mass to total leaf dry mass, exhibit a latitudinal gradient contrasting with the pattern previously observed in SLA. While LNC remains relatively constant across most latitudes, significant variations can be observed at high latitudes. Tropical and temperate forests, along with grasslands, usually show the highest LNC values, ranging from 20 to 30 $mg\ g^{-1}$. In boreal ecosystems, LNC values are typically low, often falling below 10 $mg\ g^{-1}$. LPC values tend to be moderate to low (<1$mg\ g^{-1}$) in the northern latitudes of North America, central Amazonia, some areas in central and southern Africa, central Australia, and Southeast Asia. On the contrary, high LPC (>1.7$mg\ g^{-1}$) is concentrated in Central and Eastern Asia and the Sahel, with smaller hotspots in the interior western United States and the Dry Andean Plateau, whereas Central America and the Humid Andes show few occurrences.

Trait distributions are often non-Gaussian (e.g., skewed or multimodal), so the community-weighted mean may not represent the most common value; accordingly, we compute higher moments from species abundances, standard deviation, skewness, and kurtosis, to more fully characterize functional diversity (see the Methods subsection Iterative computation of trait distribution parameters from the species abundance maps). Fig. 1b presents community-weighted standard deviation maps (CWSD) for the three traits analyzed. These maps illustrate the variability of trait values within communities, highlighting how these values diverge from the mean. Fig. 1b indicates that LNC shows the lowest standard deviation and the most stable pattern along the latitudinal gradient among the traits, with minimal variability primarily observed in central Russia and Australia. In contrast, the standard deviation maps for SLA and LPC show broadly higher variability worldwide, with pronounced hotspots in eastern Australia and the Andes; LPC variability is comparatively low across central Russia. Within Brazil, CWSD is generally lower in the core Amazon Basin than in the more seasonal and edaphically heterogeneous southeastern biomes (Cerrado and Atlantic Forest), with the most extreme values along the Amazon-Cerrado transition and in the Atlantic Forest, especially for SLA and LPC.

Our analysis reveals distinct spatial patterns in the higher-order moments of the analyzed traits (see Fig. 2), which depart from the established Gaussian assumption across several regions. In the boreal regions of North America and Eurasia, SLA and LNC show high positive skewness and excess kurtosis, indicating peaky, right-tailed distributions and low functional evenness. In tropical South America, particularly the Andes, Amazon, Cerrado, and Atlantic Forests, SLA (and often LNC and LPC) shows skewness near zero with predominantly negative excess kurtosis, consistent with flatter, more heterogeneous distributions (see previous Fig. 1b). Across Africa, LNC and LPC are predominantly flat (negative excess kurtosis). In contrast, SLA is broadly symmetric (with skewness close to zero) but exhibits greater spatial heterogeneity in kurtosis, shifting between flat and more centrally concentrated forms. Because SLA standard deviation (Fig. 1b) is generally low across the continent, these differences reflect changes in distribution shape rather than spread, pointing to regional contrasts in functional evenness. In the Sahel and surrounding drylands, LNC and LPC distributions are often left-skewed. While LNC values remain flat, LPC values are more concentrated around the mean. Elsewhere in Africa, LNC and LPC are largely flat and near-symmetric, and SLA transitions regionally between flat and more centrally concentrated

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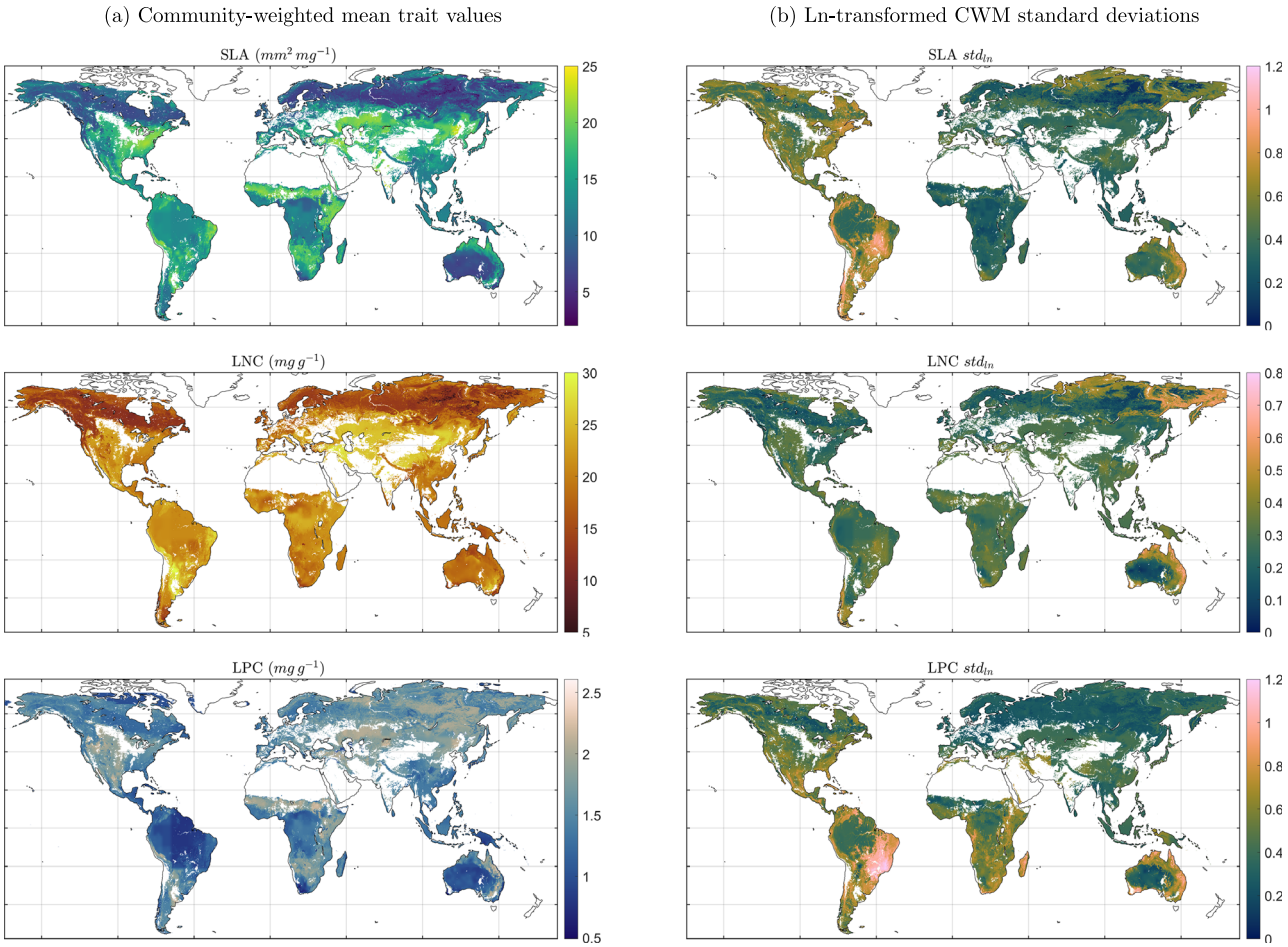


Fig. 1 | High-resolution maps of community-weighted trait means and variability. The left column **a** shows the community-weighted means (CWM), while the right column **b** displays the ln-transformed community-weighted standard deviations (CWSD) for each trait. All computations were performed in ln-transformed space (see the Methods subsection Linking species abundance to expected mean traits), but here CWM values were back-transformed to original units to enhance visual interpretability. As std is computed on the ln scale, it has a multiplicative interpretation (variation factors of $exp(\pm std)$ about the geometric mean). Weighted means and standard deviations were calculated from species abundance estimates derived from GBIF occurrence data using the proposed multiresolution approach and remote sensing-derived PFT fractional data to mitigate GBIF sampling biases.

forms. In Australia, the interior shows more pronounced left-skew values for SLA and LPC than for LNC. In contrast, the mid-latitude temperate regions of Europe and eastern North America remain mostly symmetric, with only limited deviations from flatness across traits.

Supplementary Note 1 illustrates an example of the high spatial resolution achieved by the proposed methodology in characterizing the statistical distributions of the selected traits, using the highly heterogeneous region of the Iberian Peninsula and the SLA trait as an example. Furthermore, in the Supplementary Note 1, we also show that deviations from normality in Supplementary Fig. 1 can result from mixing across PFTs. Still, mixture-like signals remain when moments are calculated within individual PFTs (Supplementary Fig. 2). Therefore, the mapped skewness and excess kurtosis appear to reflect both compositional variation and heterogeneity within PFTs rather than an exclusively mixed-Gaussian mechanism.

Including PFT fractional information in computing CWMs

An important aspect of the proposed methodology involves incorporating remote sensing data to model spatially explicit fractional PFT distributions. These datasets were derived by downscaling the MODIS land cover product (MCD12Q1, Land Cover Type 5: Annual Plant Functional Types classification) at 500 m and aggregating it to compute PFT fractions at a 1 km resolution (see Supplementary Note 9). This fractional information was introduced to mitigate biases related to spatial heterogeneity and uneven sampling, which are common in species-occurrence data from crowd-sourced databases such as GBIF. In this section, we analyze the variability of plant traits both within and among PFTs. Specifically, we compare the approach of incorporating fractional PFT information within 1 km pixels when calculating CWMs (the proposed method) to the alternative of disregarding fractional contributions by considering pure PFT classes within grid cells.

The global trait distributions indicate low normality and considerable variance, even when analyzing species assemblies within individual PFTs (Fig. 3). Comparing the trait value distributions with and without incorporating PFT fractional information to compute CWMs reveals that both methods align well in most cases when considering median values across all traits. To further investigate and visually illustrate the high spatial variability of trait values within PFTs, we analyzed the spatial distributions of LNC across Europe for each PFT, excluding the influence of PFT fractional information (Supplementary Note 2). This analysis revealed significant spatial variability driven solely by our computed species abundances from GBIF data.

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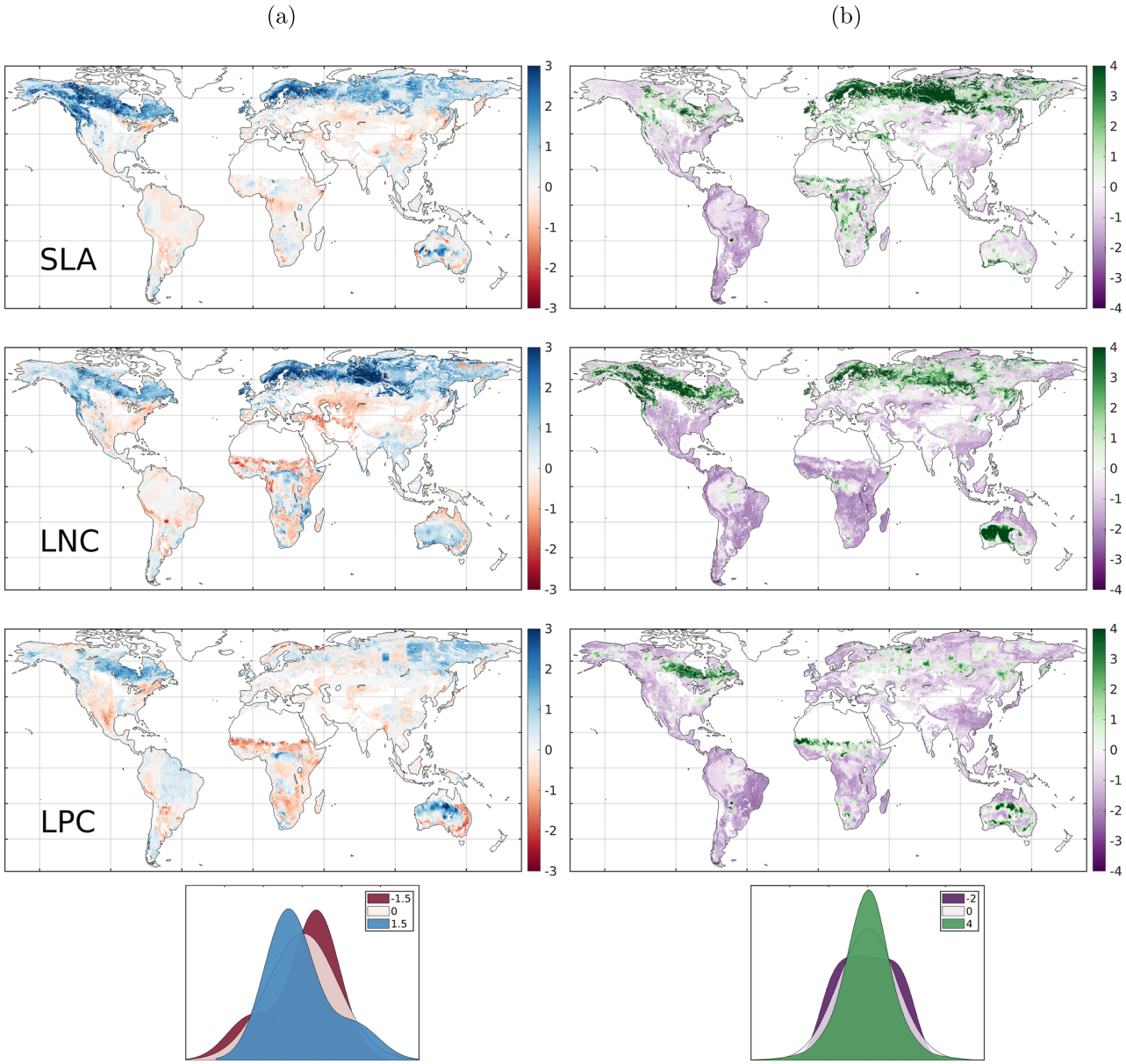


Fig. 2 | Skewness and excess of Kurtosis (*kurtosis* − 3) of the considered traits at 1 km spatial resolution. Panel **a** shows the community-weighted skewness and panel **b** the community-weighted excess kurtosis for SLA, LNC, and LPC, both capturing departures from normality in community trait distributions. Similar to CWMs, these moments were calculated from species abundance estimates derived from GBIF occurrence data using the proposed methodology.

We compared our results with reference median values reported in the literature from the TRY database[2], and observed a strong level of agreement across various PFTs and traits, particularly for the fractionally unweighted version of our PFT fractional results (see Fig. 3b, right column). This alignment was anticipated, as the TRY database[2] did not account for mixed compositions.

Discrepancies in trait distributions are observed when analyzing mixed PFT pixels during CWM computation, as shown in Fig. 3a, b, including fractional landscape data results in an increase in variance across the distribution functions. For deciduous needleleaf forests (DNF), the initial distribution, without fractional composition data, shows low variance for SLA and LNC, and a double-peaked pattern for LPC. The distributions become broader after adding fractional composition data, and the LPC double peak disappears. Additionally, distributional shifts are observed for shrubland types (SHR), with an increase in variance. In contrast, evergreen needleleaf forests (ENF) show minimal changes when fractional data is included.

Comparison with the sPlotOpen database

The sPlotOpen database[19] represents the most comprehensive open-access resource for abundance-based plot trait estimates currently available. It is important to clarify that these trait values were not directly measured in the vegetation plots but instead derived from the gap-filled version of the TRY database[18]. While we provide our estimates at a constant 1 km spatial resolution, the sPlotOpen database is composed of a heterogeneous compilation of plots of different sizes and unevenly distributed around the planet (see Methods). To ensure robust comparisons, we restricted our analysis to plot locations with trait coverage above 0.8, plot size to 100–10,000 m^2, and location uncertainty to ≤300 m. Additional details on how plot size and trait

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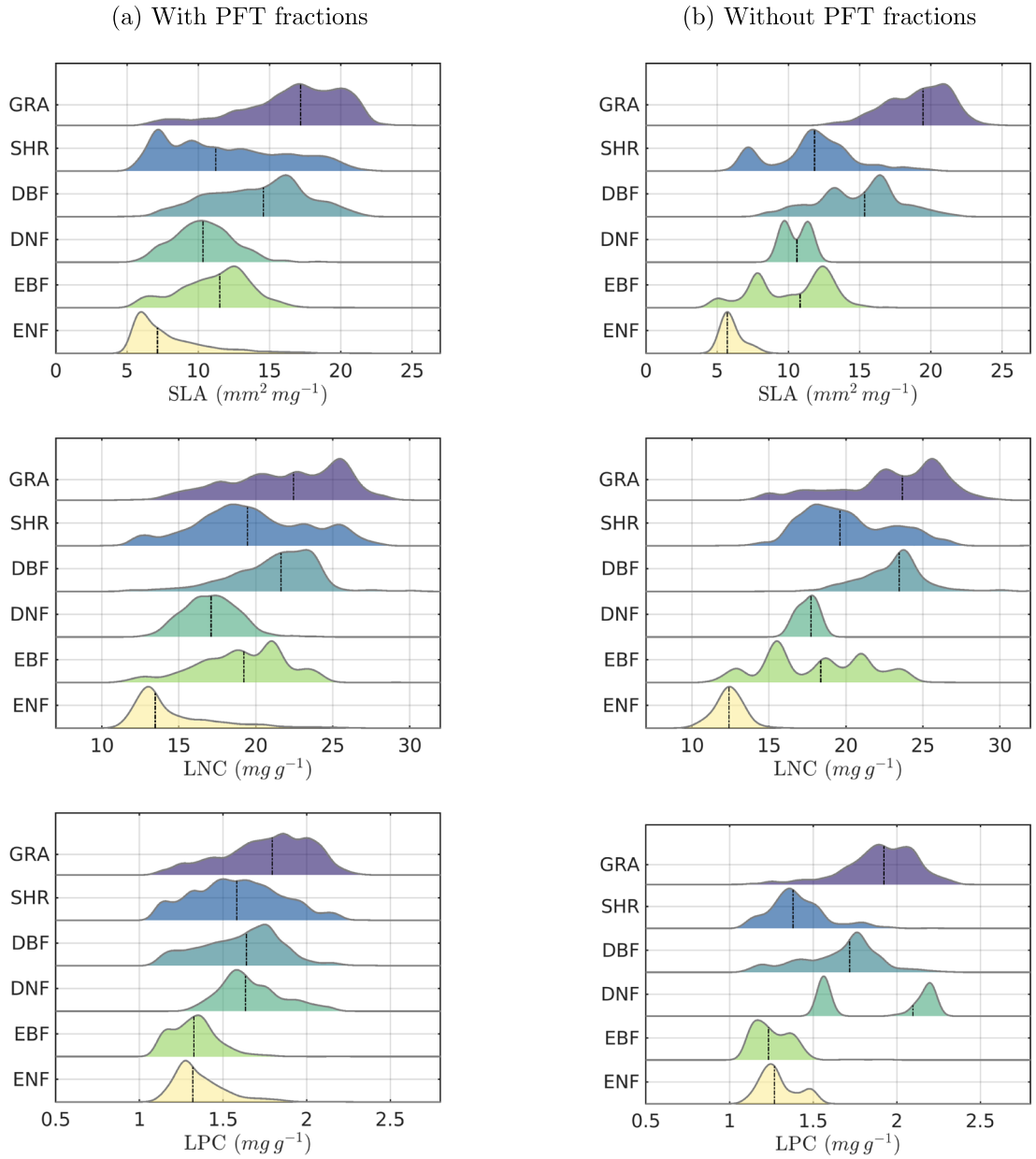


Fig. 3 | Density distributions of trait values by dominant plant functional type (PFT), with PFTs distinguished by color. All computations were performed in ln-transformed space, but CWM values were back-transformed to original units to enhance interpretability. In the left column, CWMs are calculated based on the fractional composition of each 1 km pixel (the proposed approach), which accounts for both between- and within-PFT variability. In contrast, the right column assumes each pixel is uniformly represented by its dominant PFT, capturing only within-PFT variability. ENF refers to evergreen needleleaf forest, EBF for evergreen broadleaf forest, DBF for deciduous broadleaf forest, DNF for deciduous needleleaf forest, SHR for shrubland, and GRA for grassland.

coverage influenced the residuals observed in our comparisons, along with the rationale for these thresholds, are provided in Supplementary Note 3.

In alignment with recent works[20], we calculated top-of-canopy weighted mean (TWM) trait estimates for each plot as a complement to the CWM values provided by sPlotOpen, which integrate traits across all recorded vegetation layers. Because these alternative community definitions can differ substantially, they can lead to marked differences in plot-level trait estimates (see Supplementary Note 4). We derive TWMs by weighting plot species lists using remotely sensed growth-form fractions (trees, shrubs, and grasses), thereby emphasizing dominant PFTs (see Vegetation plot data in the Methods for further details). This canopy-oriented comparator is particularly relevant for optical remote sensing, which is primarily sensitive to top-of-canopy properties. Accordingly, our 1-km products are best interpreted as canopy-biased community means rather than exhaustive all-strata plot CWMs. We therefore benchmark against (i) sPlotOpen plot-level CWMs as an open all-strata reference and (ii) TWM as a complementary, canopy-aligned comparator. Since TWM uses remotely sensed growth-form fractions as weights, it is reported here as a complementary benchmark (sensitivity to community definition). In Fig. 4, the x-axis shows our 1km predictions extracted for sPlotOpen plot locations, and the y-axis shows the corresponding sPlotOpen plot-level trait (CWM in the left panels; TWM in the right panels). Using TWM trait estimates from sPlotOpen data yields a more uniform distribution along the 1:1 line compared with our estimates, indicating a reduced residual spread (see Fig. 4). For SLA and LNC, the slopes between sPlotOpen and our estimates approach one (red line), signifying stronger agreement

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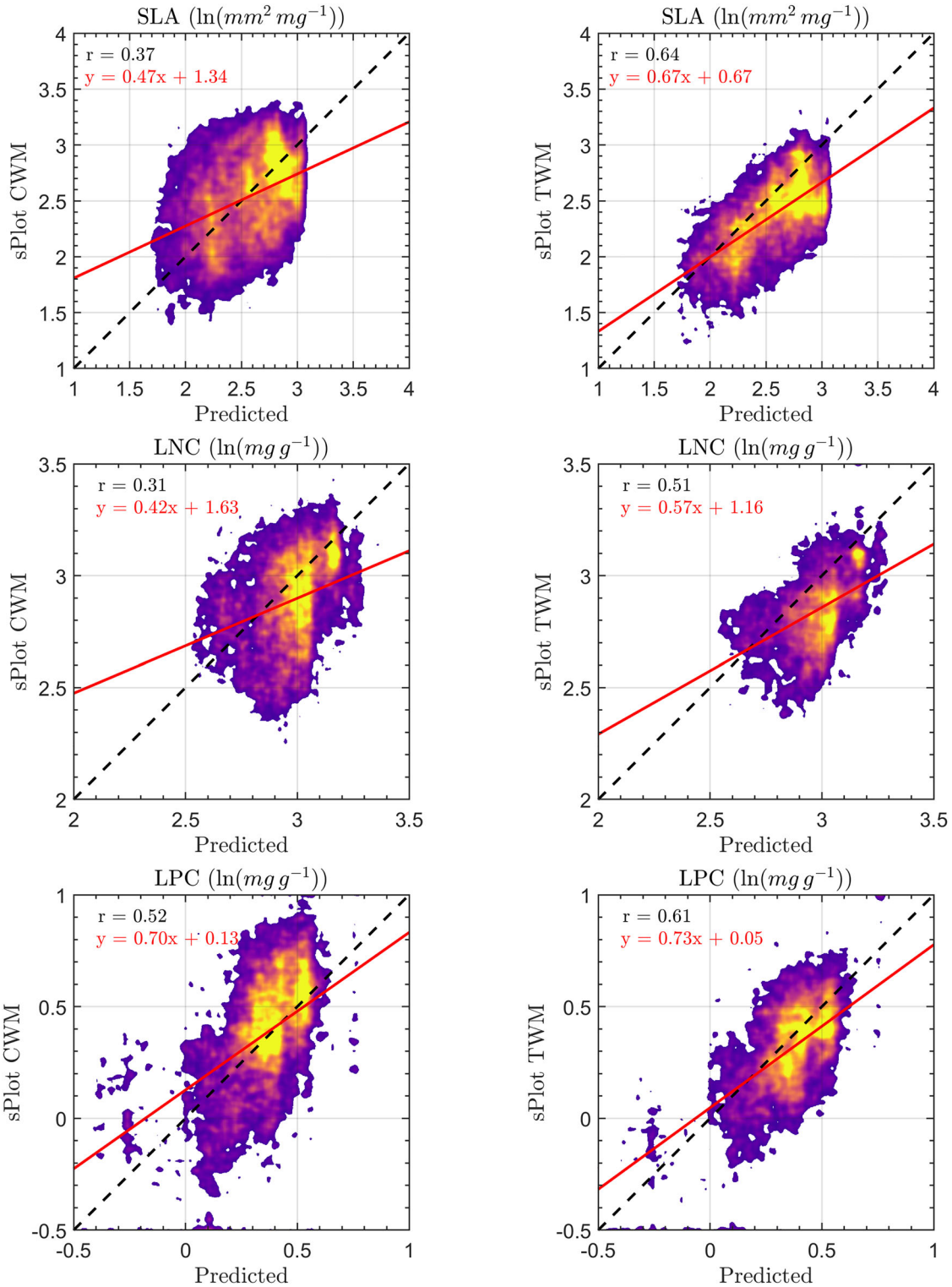


Fig. 4 | Scatterplots comparing our ln-transformed 1-km trait estimates (*x* axis) with sPlotOpen plot-level traits (*y* axis) at matched plot locations. Left column: sPlotOpen community-weighted means (CWM). Right column: sPlotOpen top-of-canopy weighted means (TWM) computed from the plot-specific species lists and weighted by remotely sensed growth-form fractions (trees, shrubs, grasses). Because the TWM weights rely on remotely sensed growth-form fractions, the TWM panels are intended as a canopy-aligned benchmark. Black dashed line = 1:1; red line = least-squares fit. Pearson's correlation tests (*r*) are two-sided (all $p < 0.001$). Point color encodes local point density: purple indicates lower density, orange moderate, and yellow/white higher. Comparisons include only plots restricting plot size to 100–10,000 m² and location uncertainty to ≤300 m (see Supplementary Note 3 and Methods).

between datasets. Supplementary Fig. 6 further examines residual variation by growth-form fractions, showing a consistently lower spread (reduced RMSE) and no clear dependence on fractions compared to the CWM sPlotOpen counterpart.

Comparisons with sPlotOpen show that TWM consistently outperforms CWM on both the *ln*-transformed and original-unit scales (Table 1). Metrics are computed on the *ln* scale to align with sPlotOpen and to better approximate a normal distribution[2,19,22]. For readability,

Table 1 | Agreement with sPlotOpen by trait and community definition

Scale (units)	Trait	Group	Intercept	Slope	*r*	R^2	RMSE	Rel. RMSE (%)
(A) ln-transformed								
ln($mm^2\ mg^{-1}$)	SLA	CWM	1.34	0.47	0.37	0.14	0.43	18.03
ln($mm^2\ mg^{-1}$)	SLA	TWM	**0.67**	**0.67**	**0.64**	**0.38**	**0.36**	**14.98**
ln($mg\ g^{-1}$)	LNC	CWM	1.63	0.42	0.31	0.10	0.25	18.12
ln($mg\ g^{-1}$)	LNC	TWM	**1.16**	**0.57**	**0.47**	**0.22**	**0.23**	**16.81**
ln($mg\ g^{-1}$)	LPC	CWM	0.13	0.70	0.52	0.27	0.26	14.52
ln($mg\ g^{-1}$)	LPC	TWM	**0.05**	**0.73**	**0.61**	**0.38**	**0.22**	**11.82**
(B) original units								
$mm^2\ mg^{-1}$	SLA	CWM	7.64	0.44	0.34	0.12	5.54	19.07
$mm^2\ mg^{-1}$	SLA	TWM	**3.94**	**0.55**	**0.56**	**0.32**	**4.28**	**14.74**
$mg\ g^{-1}$	LNC	CWM	10.03	0.42	0.33	0.11	4.32	19.63
$mg\ g^{-1}$	LNC	TWM	**6.71**	**0.55**	**0.48**	**0.23**	**4.02**	**18.22**
$mg\ g^{-1}$	LPC	CWM	0.42	0.76	0.50	0.25	0.37	16.06
$mg\ g^{-1}$	LPC	TWM	**0.36**	0.71	**0.59**	**0.35**	**0.28**	**12.25**

For each trait and group (CWM/TWM), we report the group-specific intercept and slope (from the ANCOVA[59]; simple fit per group, further details in Supplementary Note 5), Pearson's *r*, R^2, RMSE, and Relative RMSE (%). All statistical tests are two-sided. Boldface highlights TWM values that improve over the corresponding CWM under the following criteria: intercept closer to 0, slope closer to 1, larger r/R^2, and smaller RMSE/Rel. RMSE.

Intercept and slope are the group-specific coefficients implied by the ANCOVA (baseline plus group shift/interaction). RMSE and Relative RMSE (%) are reported on the stated scale; Rel. RMSE corresponds to the normalized percentage metric used in our evaluation. Rel. RMSE (%) is computed as $100 \times \mathrm{RMSE}/(Q_{0.99} - Q_{0.01})$ on the stated scale (reference quantiles), and is not a multiplicative percent error. For a multiplicative interpretation of ln-RMSE σ, a 95% factor-of-error band on the back-transformed scale is $\exp(\pm 1.96\sigma)$. Using table values (two decimals): SLA (CWM 0.43: 0.43–2.32×; TWM 0.36: 0.49–2.03×); LNC (CWM 0.25: 0.61–1.63×; TWM 0.23: 0.64–1.57×); LPC (CWM 0.26: 0.60–1.66×; TWM 0.22: 0.65–1.54×). All highlighted TWM improvements are statistically significant at the level supported by the data and tests described in Supplementary Note 5.

we also report back-transformed (original-unit) values with their multiplicative interpretations. Under CWM, association was weak-to-modest on the ln scale (r = 0.31–0.52, R^2 = 0.10–0.27) with ln-RMSE = 0.25–0.43 and relative ln-RMSE = 14.52 − − 18.12%. Here, Rel. RMSE (%) denotes RMSE relative to the 1st–99th percentile range of the reference values, Rel. RMSE(%) = $100 \times \mathrm{RMSE}/(Q_{0.99} - Q_{0.01})$ (computed on the same scale as RMSE), which provides a robust scale term for cross-trait comparability. Switching to TWM increased association (r = 0.47–0.64, R^2 = 0.22–0.38), improved calibration (slopes 0.57–0.73 vs. 0.42–0.70; intercepts closer to 0: SLA 1.34 → 0.67, LNC 1.63 → 1.16, LPC 0.13 → 0.05), and reduced ln-RMSE (SLA 0.43 → 0.36, LNC 0.25 → 0.23, LPC 0.26 → 0.22) and relative ln-RMSE (SLA 18.03 → 14.98%, LNC 18.12 → 16.81%, LPC 14.52 → 11.82%). In original units, RMSE and relative RMSE also declined (SLA 5.54 → 4.28 and 19.07 → 14.74%; LNC 4.32 → 4.02 and 19.63 → 18.22%; LPC 0.37 → 0.28 and 16.06 → 12.25%). Interpreted multiplicatively, the TWM ln-RMSE of 0.36 for SLA corresponds to a 95% factor-of-error of ≈ 0.49–2.03×, with similar narrowing for LNC and LPC.

These quantitative improvements are consistent with results shown in Supplementary Note 5, where paired comparisons on the same plots indicate that Δ = TWM − CWM < 0 across traits and different tested remotely sensed fractional products, and that calibration and association improve when using TWM. The same pattern is visible qualitatively in Supplementary Note 4: violin plots of residuals by target and scatter plots contrasting sPlotOpen CWM with sPlotOpen TWM display a systematic downward shift for TWM and a tighter spread of residuals. Because spatial aggregation can mask sub-km heterogeneity, we also examined whether performance changes depend on the grid cell size used to compute abundances from the proposed multiresolution approach. Supplementary Note 8 includes maps of the optimal sampling quadrat size selected by our multi-resolution procedure and stratifies residuals by this size. Residual medians remain near zero and show no monotonic trend across size bins, indicating that the grid cell size used for abundance estimation does not systematically drive gains or losses.

Finally, we adjusted the grid cell size to 0.5° to compare our results with the coarser spatial resolution of other occurrence-based methods[28] (see Supplementary Note 6). Our method consistently outperformed theirs, achieving higher correlations for SLA, LPC, and LNC. Specifically, the SLA correlation increased from 0.31 to 0.38, the LNC from 0.30 to 0.44, and the LPC from 0.35 to 0.38. Furthermore, our approach significantly reduced estimation biases. While absolute agreement remains modest at this coarse-grained level, these gains are consistent in direction across traits and benchmarks. We therefore interpret the 0.5° comparison primarily as a like-for-like benchmark against occurrence-only upscaling at the same resolution, rather than as evidence of plot-scale predictive accuracy. These improvements suggest that the proposed refinements can partially mitigate the effects of sampling biases inherent in simplistic occurrence-based methods[28]. In line with recent global intercomparisons, modest metrics at this scale are not unique to our approach but reflect shared constraints of global trait-map benchmarking[20].

On the representativeness of the TRY database

To assess consistency, we compared species coverage in the TRY database, which provides trait data, with the more comprehensive GBIF database, used here as a reference for global species representation. This comparison highlights gaps in trait data for species present in GBIF but absent in TRY, helping to potentially identify regions with greater uncertainties and set priorities for future sampling to better capture global trait diversity. We accounted for these missing species by weighting their contribution based on computed abundances, emphasizing their significance at a grid cell level.

Figure 5 illustrates species abundance in GBIF but absent from TRY, using LNC as an example, as other traits show similar spatial patterns (see Supplementary Fig. 7 for SLA and LPC). The Northern Hemisphere better represents traits, with <40% abundance of species unrepresented, particularly in Europe, Canada, and the US, where TRY observations are more extensive[18]. However, grassland and shrubland areas in the US, Central Asia, and China exhibit lower representation, likely reflecting higher species diversity and weaker trait coverage in these biomes. Trait representation declines significantly in equatorial and southern latitudes, with darker red areas in Fig. 5 marking the most trait data-deficient regions. In highly diverse tropical regions

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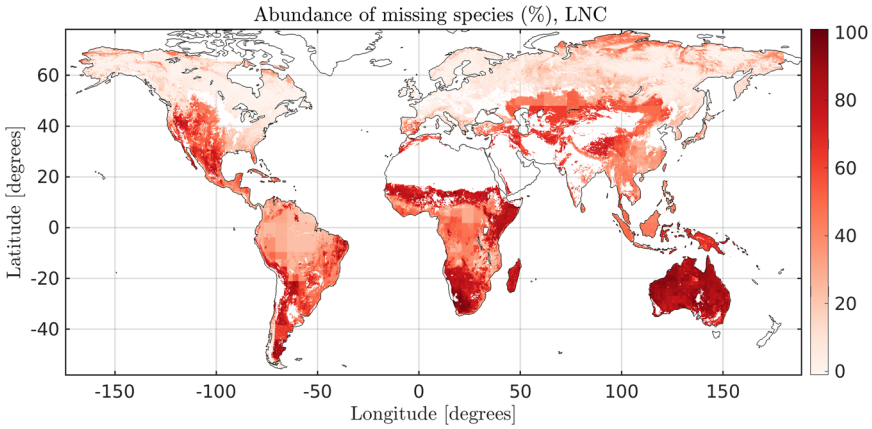


Fig. 5 | Abundance-weighted trait data deficiency for LNC. Spatial distribution of the percentage of GBIF-estimated community abundance associated with species lacking trait data in the TRY database. Higher values could indicate regions of elevated uncertainty in trait estimation.

(including parts of the Amazon and Congo basins), TRY–GBIF representativeness indicates that a larger fraction of the GBIF-estimated community abundance is associated with species lacking trait data and/or with limited cross-database linkage. These areas should therefore be interpreted jointly with the representativeness layers as regions of elevated uncertainty. Supplementary Note 8 provides a complementary diagnostic of occurrence-data limitations: meeting the sampling-completeness threshold in the Amazon often requires larger effective quadrats, implying that broader spatial support is needed to accumulate sufficient records to characterize locally dominant assemblages. Grasslands and shrublands exhibit the highest percentage of missing species, with over 80% unaccounted for in TRY. Africa and Australia are the most poorly represented regions overall, though gaps vary by trait (Supplementary Fig. 7).

Discussion

We introduce a multiresolution approach that utilizes global trait and species-occurrence datasets with remote-sensing data to create high-resolution maps of plant trait distributional descriptors, departing from conventional statistical and machine-learning modeling approaches. At broad scales, observed gradients in SLA, LNC and LPC follow classic trait-environment trade-offs (leaf economics spectrum): warmer or fertile sites favor acquisitive strategies, whereas cold or water-limited systems select conservative leaves[32,33].

According to recent intercomparison studies, our methodology outperforms previous methods in the literature that focused solely on remote sensing data[20]. This improvement is even more noticeable when viewing the enhanced spatial resolution of our estimates and the reduction in spatial aggregation, which was a common practice in previous comparison studies[20]. However, the accuracy of comparisons with sPlotOpen is inherently constrained by limitations in the reference dataset itself[19]. These limitations include its non-representative sampling design, geographic biases from opportunistic data collection, and inconsistencies in plot sizes or taxonomic scope (e.g., exclusive focus on woody species in some tropical plots). Such factors introduce uncertainties that complicate direct validation and suggest that even future methodologies may not achieve higher agreement with sPlotOpen under current data constraints. The limited availability of open, globally consistent datasets providing community-weighted trait means (and higher moments) beyond sPlotOpen limits direct benchmarking, particularly in undersampled regions. Developing and harmonizing such resources into globally accessible validation benchmarks should be a priority for future, region-focused evaluations. Finally, although TRY supplies species-level trait means for both our mapping and the sPlotOpen used for evaluation, an unavoidable, field-wide circularity[20], its impact on our comparisons should be limited, as concordance is driven primarily by independently derived species-abundance patterns obtained from different data sources and methods.

We compared our results with those proposed by ref. 28, which rely entirely on averaging biodiversity citizen science data. While their approach generates trait estimates comparable to those of sPlotOpen, it differs in scope and spatial resolution. Our method, however, demonstrated superior performance in both spatial resolution and agreement with sPlotOpen reference data across all the analyzed traits. The spatial aggregation process employed by ref. 28 (highest resolution ~11 km^2) effectively reduces noise variance by averaging partially noisy estimates, thereby minimizing random fluctuations and outliers and improving the signal-to-noise ratio. While this enhances statistical robustness, it also blends distinct PFTs, resulting in oversimplified trait distributions that obscure ecological complexity and fine-scale variation in vegetation structure and function[20].

Similarly, previous works[20,21] showed that models relying solely on climatic data often yield overly smooth, macro-environmentally dominated patterns. While such smoothing robustly captures broad gradients, it can obscure ecologically meaningful variation associated with microhabitats, soil heterogeneity, topography, land use, and microclimates. To partially address this trade-off, we integrate remotely sensed PFT fractional composition to reintroduce sub-kilometer heterogeneity that GBIF-derived abundances alone cannot resolve due to spatial aggregation. We emphasize, however, that fine-scale remote-sensing layers can carry their own noise and biases (e.g., classification and geolocation errors, mixed pixels), whereas broad climatic fields are comparatively stable. In our framework, PFT fractions are used as a complementary constraint intended to refine, rather than replace, the broad climatic signal and the patterns captured by GBIF-derived abundances. This balanced integration is designed to retain large-scale gradients while adding spatially explicit, ecologically plausible heterogeneity to the trait maps and, consistent with[35], may support a more nuanced representation of vegetation structure and function.

The calculated trait maps demonstrate unique spatial patterns that are not necessarily linked to the included PFT data. This suggests we can capture well-known trait differences between PFTs and significant variations within PFTs due to contrasting species distributions and community assemblages. Combining these characteristics enables us to overcome some of the limitations of previous PFT-based approaches, which also produced maps of plant traits at large scales[20]. Moreover, our method identifies areas with insufficient trait data, directing future sampling strategies by analyzing species coverages from the TRY and GBIF databases. It reveals notable data

deficiencies, particularly in equatorial and southern regions, as well as in grassland and shrubland ecosystems, highlighting the need for targeted trait sampling in these locations.

We evaluated the impact of discrepancies between CWMs and TWMs on trait estimation, a key factor influencing modeling approaches. Using TWMs from sPlotOpen as a canopy-oriented comparator, our method showed a significant increase in explained variance and reduced bias by incorporating fine-scale (1 km^2) fractional growth-form composition. This is ecologically expected, as the optical sensor information used in this work emphasizes sunlit canopy strata, where trait syndromes differ from shaded understory layers[20]. On the same plots, Δ = TWM − CWM is consistently negative, and sPlotOpen CWM vs. TWM are statistically distinguishable for nearly all trait-product combinations. This inferential result is also consistent with qualitative diagnostics and supports using TWM, which is closer to the top-of-canopy signal sensed by optical data, as a complementary, canopy-aligned comparator for our approach, as it yields lower RMSE, better calibration (slopes/intercepts nearer (1, 0)), and higher R^2. However, because TWM weights are derived from remotely sensed growth-form fractions (and global fractional products may share upstream sensor inputs), this comparison is not fully independent and should be interpreted as a sensitivity analysis to community definition. Residual discrepancies, especially for LPC, are primarily driven by intercept shifts and likely result from differences in spatial aggregation, vertical layering, and within-growth-form mixtures. This high-resolution approach captured top-of-canopy variations, likely reflecting microclimatic influences in sPlotOpen. However, we identified significant discrepancies between the growth form fractions reported in sPlotOpen and those estimated using remote sensing, impacting species abundance estimates. These differences stem from two main factors. First, growth form representation varies as sPlotOpen includes understory vegetation while remote sensing primarily captures top-of-canopy layers. Second, measurement errors in abundance, geolocation uncertainties, and variations in plot size in sPlotOpen introduce inconsistencies, which could be particularly relevant in heterogeneous landscapes at fine spatial resolutions.

Averaging trait estimates over 1 km grid cells can introduce aggregation-related biases, including the Modifiable Areal Unit Problem (MAUP) and neighborhood effects[36]. To limit these, we (i) estimate species abundances with a multiresolution procedure that selects the smallest quadrat achieving sample coverage ≥0.8 and discards undersampled units, and (ii) propagate sub-kilometer heterogeneity into grid cell trait moments using fractional PFT cover from Landsat-informed downscaling of MODIS rather than majority classes. Because the workflow relies on empirically estimated abundances rather than a fully parametric model, closed-form uncertainty propagation is not available. Instead, we provide two spatial diagnostics, the selected optimal quadrat size (Supplementary Note 8) and an abundance-weighted TRY–GBIF representativeness layer (see Results), which indicate where aggregation and sampling limitations likely elevate errors. Consistently, analysis of the residuals stratified by quadrat size shows no systematic drift (Supplementary Fig. 9). Additionally, classification uncertainty in the MODIS-to-Landsat downscaling remains a global (non-mapped) source of uncertainty. Finally, in addition to spatial bias corrections[37], future iterations should incorporate cross-scale upscaling/downscaling that explicitly links fine- and coarse-resolution variables, potentially improving ecological fidelity and coherence across resolutions.

We estimated ecosystem-level traits and characterized their within-cell distributions using higher moments (standard deviation, skewness, excess kurtosis). Our results challenge the assumption that plant trait distributions are lognormal (i.e., Gaussian on the natural-log scale). Even after natural-log (ln) transformation, traits exhibit pronounced skewness and kurtosis in broad areas of the planet, indicating that higher statistical moments are needed to capture ecosystem functional diversity. Thus, beyond means, skewness and kurtosis carry ecological signal: right-skew can reflect strong environmental filtering or recent disturbance, whereas flatter distributions suggest greater functional evenness and potential niche complementarity[38]. Although our 1 km products do not resolve neighborhood-scale processes underlying species coexistence, they provide a view of landscape-scale variation in community trait distributions (Supplementary Figs. 1 and 2). In this study, we refer to coexistence-aware models as trait-based community and vegetation models that represent competition and population dynamics as functions of both species' trait values and the degree of trait variation within the community. In this context, our gridded trait distributions can help to generate hypotheses and can be used as spatial priors or constraints, or as benchmarking targets to test whether models reproduce observed patterns in both trait means and dispersion, but they should not be interpreted as direct evidence of interaction mechanisms.

Consistent spatial variation in trait distributions provides quantitative baselines for biodiversity monitoring and enables patch-specific restoration targets, including trait means, dispersion, and functional evenness, to guide species selection and abundance allocation for maintaining ecosystem multifunctionality[38]. In this context, prior studies[12] demonstrated that fully accounting for leaf functional diversity, including its non-Gaussian aspects, significantly influences modeled carbon fluxes. This emphasizes that incorporating the complete trait distribution information is crucial for accurately predicting ecosystem productivity. While direct validation is not yet feasible, these maps provide a benchmark for future assessments and methodological refinements as new reference data emerge. The higher-moment layers, in particular, are offered as proof-of-concept products and should be interpreted with caution, as no global standards exist for comparison. We therefore recommend consulting the accompanying spatial diagnostics to assess local reliability and to flag regions where aggregation or sampling limitations are likely to inflate uncertainty, especially across sharp soil or disturbance transitions where fine-scale heterogeneity is highest. Framed this way, the products can guide hypothesis generation and help prioritize targeted data collection for subsequent validation.

Global Carbon Models, Dynamic Global Vegetation Models, and vegetation Radiative Transfer Models involve a complex interplay of physical, chemical, and biological processes across different temporal and spatial scales. Despite their utility, these models are often constrained by insufficient or imprecise data, which limits their capacity to accurately represent critical processes. This challenge hampers the identification of causal mechanisms underlying observed phenomena and undermines the reliability of parameterization efforts. An example of this problem is using vague plant trait priors in model parameterization, which can lead to obtaining equally valid solutions (referred to as equifinality problems) or generating mathematically plausible model outputs based on incorrect assumptions[39]. This highlights the importance of accurately determining trait higher statistical moments, which is necessary for adequately describing and incorporating trait information into ecosystem models[13,14].

Our approach represents a significant advance because it does not rely solely on direct correlations between spectra and individual leaf traits, which have traditionally constrained remote sensing studies. Such spectral-trait correlations tend to emphasize photosynthetic tissues and chemical properties, limiting the range of traits that can be analyzed. By contrast, our framework utilizes PFT composition and observational datasets, enabling trait estimation to be extended beyond leaf-level chemistry. Trait values are assigned from trait databases at the species level and aggregated using observation-informed abundances. In this context, remote sensing primarily serves as a structural constraint on canopy dominance, functioning as a top-of-canopy weighting rather than as a direct sensor of each trait. Consequently, extensions to woody or belowground dimensions are feasible

in principle, but should be interpreted as canopy-weighted, dominance-oriented inferences. These inferences are generally more robust for size- and structure-aligned traits, and more uncertain for root economic or hydraulic traits that may be partially decoupled from canopy signals. Previous global trait mapping studies[26,28] demonstrate the potential of this composition-driven strategy and stress the significance of clearly communicating its scope and limitations when interpreting spatial patterns of vegetation structure and function. Furthermore, as a long-term goal, alongside improving modeling techniques, future developments should include methods to generate estimates of species traits under different climate conditions, replacing the relatively simple species-mean approach used in this work and most previous methods. This enhancement would refine ecosystem trait estimates and higher-order statistical moments by incorporating increased intraspecific variance, offering a more realistic representation of plant acclimation processes.

Methods

Plant trait data

The TRY database is a comprehensive global archive of plant traits compiled through a collaborative effort between researchers and data contributors worldwide[2,18]. It is a valuable resource for ecological research and biodiversity conservation, containing data on over 250,000 plant species and their functional traits. The database encompasses a wide range of plant traits, including morphological, physiological, reproductive, and phenological characteristics. TRY aims to promote a deeper understanding of the ecological and evolutionary drivers of plant trait variation, ultimately facilitating more informed predictions of the impacts of environmental change on plant communities and ecosystems. It is important to note that TRY aggregates measurements produced under heterogeneous laboratory protocols. Methodological differences across contributors, particularly for LNC, SLA, and LPC used in the present work, likely introduce residual variance. We applied ln-transformations to attenuate these effects, but some artifacts may remain.

Species-occurrence data

The GBIF is the largest open-access database providing free access to occurrence data for plants, animals, and other organisms[34]. GBIF collects, integrates, and shares biodiversity data from various sources, including museums, research institutions, government agencies, and citizen scientists. The database is curated and maintained by a global network of institutions. It contains over one billion records of plant species distributions and geolocated occurrences from all over the world. The database provides a crucial resource for scientists and researchers interested in studying the distribution and ecology of vascular plant species at both regional and global scales.

The present study relies on geolocated occurrence records extracted from the GBIF database. Figure 6 illustrates the global spatial distribution of these records and shows an apparent disparity in the density of occurrence records across different regions. More specifically, Europe, Australia, and the Contiguous United States stand out as regions with substantial data availability. In contrast, other parts of the world, such as Africa, South America, and Asia, show lower densities of occurrence records. In addition, the observed differences in the density of occurrence records highlight the need for more comprehensive data collection efforts in regions where information is too sparse or lacking. In the following sections, we will introduce the developed methods for compensating for the uneven distribution of occurrence records, thereby enhancing the reliability of the abundance data extracted from GBIF.

Vegetation plot data

The sPlotOpen dataset, a subset of the sPlot vegetation plots dataset, contains 95,000 open-access records. It was assembled for environmental and spatial balance, yet it still shows uneven geographic coverage, with some regions and biomes sparsely represented[19]. Together with compositional information, sPlotOpen also provides ln-transformed CWMs for most traits. It is important to note that the trait values in sPlotOpen were not measured on-site in the vegetation plots; instead, they were obtained from the gap-filled version of the TRY database[18] and matched to species names to calculate a community-weighted mean per plot and trait[40].

We used the plot-level CWMs provided by sPlotOpen and derived TWM estimates for each plot to assess the influence of vertical stratification on foliar traits. To derive TWMs from sPlotOpen, we computed growth-form-specific community-weighted means (CWMs; trees, shrubs, grasses) within each plot from the sPlotOpen dataset, and then derived TWM by combining these CWMs with the remotely sensed growth-form fractional covers at 1km spatial resolution for each plot. Specifically, with ENF, EBF, DNF, and DBF aggregated as "trees," we define

$$\mathrm{TWM} = \sum_{g \in \{\text{trees, shrubs, grasses}\}} f_g \, \mathrm{CWM}_g, \tag{1}$$

where f_g are top-of-canopy-sensitive fractional growth-form covers estimated from our PFT map (Supplementary Note 9). This statistic highlights the overstory trait signal that optical sensors capture, providing a like-for-like comparator to our remote sensing-informed trait estimates. To ensure robustness, we replicated the TWM computation using two independent high-resolution land cover products (GLC_FCS30D[41] and ESA WorldCover 10 m[42]), yielding consistent results (Supplementary Note 5).

Additionally, we used geolocation data (latitude/longitude) from sPlotOpen to align plot locations with our pixel estimates and to incorporate ancillary information for each vegetation plot, including plot area, location uncertainty, and trait coverage. This integration enabled a more detailed analysis of potential factors influencing deviations observed in the sPlotOpen data.

Computing abundance maps from biased species-occurrence data

Accurate computation of trait CWMs requires quantitative measurements of the relative abundances of plant species in the ecosystem[43]. Nevertheless, comprehensive global abundance estimates do not exist, and available data are typically limited to the site or plot level[19,44]. Large datasets, such as the GBIF, primarily provide species-occurrence data, which poses challenges for estimating abundances. GBIF's inconsistent design and sampling schemes introduce significant spatial biases in species distribution data[45], potentially undermining efforts to model species distributions. Similarly, citizen science datasets such as iNaturalist (part of GBIF) can capture broad global gradients in plant traits but exhibit substantial sampling biases. These biases negatively affect community-level trait estimates, particularly at finer spatial resolutions[26,28].

Thus, estimating abundance from presence data is a crucial and unresolved step for tracking biodiversity changes or computing the CWM of plant traits. Several methods have been developed to estimate species abundance from presence data[29,30]. These include regression approaches[46], methods based on counting occupied cells of a rectangular grid[47], zero-inflated count models[48], and model count data[49]. However, these methods have limitations, such as requiring knowledge of the average area occupied per individual of a species, and difficulties in collecting count data. Therefore, estimating species abundances from occurrence data, particularly at broader scales, remains a challenging and unresolved problem.

This study presents a computationally efficient method to estimate species abundances using occurrence databases such as GBIF, employing a three-step refinement process to generate global

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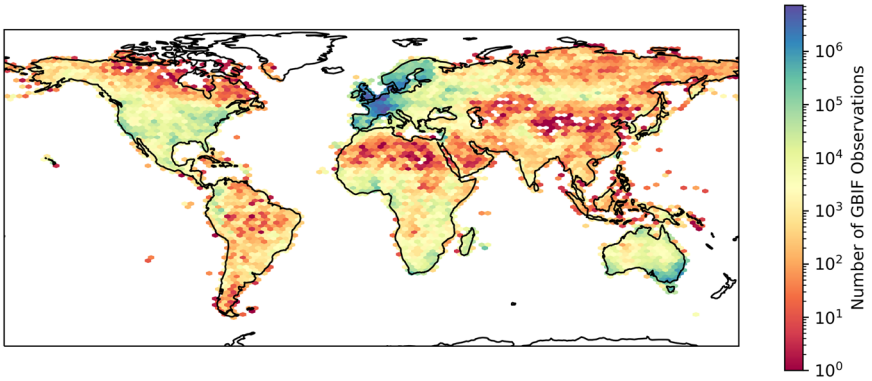


Fig. 6 | Spatial distribution of the number of geolocated plant occurrence observations in the GBIF database. The map represents the density of GBIF plant occurrence records aggregated into 100 km grid cells displayed on a logarithmic scale, illustrating the several orders-of-magnitude difference in sampling effort across regions.

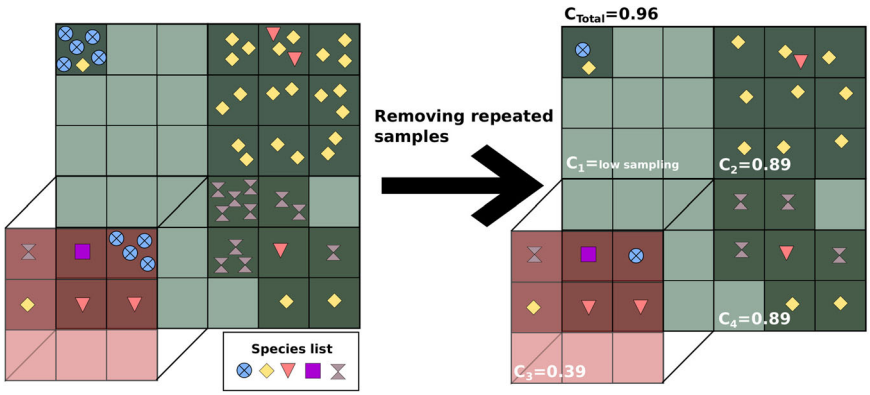


Fig. 7 | Multiresolution approach for computing species abundances. A visual example of estimating abundances from occurrence data and selecting the optimal sampling quadrat side size (S) based on sample completeness (C). In the example, $S = ISQS \cdot 2^n$, where $n \in \mathbb{Z}$, $0 \le n \le 1$, and ISQS represents the initial sampling quadrat size (which is 3 in this particular case). The reddish sampling quadrat ($S = 3$) indicates an example of the original ISQS. In contrast, the greenish sampling quadrat shows a bigger sampling quadrat ($S = 6$). Shaded grid cells represent occupied cells sampled at least once, and C_i are the completeness values computed with equation (2). According to the completeness computations of each sampling quadrat, 2 ($C_2 = 0.89$) and 4 ($C_4 = 0.89$) will use the computed abundances because their completeness values surpass the prescribed threshold ($C_i \ge 0.80$). Alternatively, 1 and 3 will consider the computed abundance values from the bigger sampling quadrat with higher completeness ($C_{Total} = 0.96$). This is because data in 1 does not have enough sampled grid cells, and data in 3 has a completeness ($C_3 = 0.39$) value lower than 0.8.

abundance maps at a 1 km^2 spatial resolution. The first step calculates abundances separately for each PFT across a gradient of sampling quadrat sizes (S), ranging from 40 km^2 to 3 × 10^3 km^2. This range is selected as a trade-off between spatial resolution and computational efficiency, considering the considerable number of cells that need to be processed per species at finer resolutions. In the second step, a multiresolution approach is applied to adaptively determine the optimal sampling quadrat size for each location, prioritizing the inclusion of the most dominant species. Finally, a remote-sensing-derived fractional PFT map is integrated to correct and refine the computed abundances, mitigating potential biases introduced in earlier steps while enhancing spatial detail and capturing the fine-scale composition of vegetation.

Estimating species abundance data from occurrence data

As an initial filtering step, all species classified as aquatic or semi-aquatic were excluded to focus exclusively on terrestrial plant diversity. Additionally, we preprocessed the raw GBIF data using the `CoordinateCleaner` R package[50], which provides efficient and standardized functions for cleaning occurrence data in ecological, biogeographic, and paleontological research.

We implemented a relatively operational but straightforward grid cell counting method, commonly used in biodiversity research[51], to calculate abundances based on unevenly collected plant species presence data from GBIF. This method partitions the study area into sampling units or sampling quadrats (S) of specified dimensions (e.g., 40 × 40 km), each further subdivided into smaller quadrats (e.g., 5 × 5 km). The frequency of species presence was recorded in each quadrat, with duplicates only counted once. It was considered absent if a species was not registered in a given sampling quadrat. The sum of all frequencies across small quadrats within a 40 × 40 km sampling unit represented the species abundance. For example, a species' highest abundance within a 40 × 40 km unit is one, indicating that it is present in all possible sampled quadrats at least once. Abundance in this context is explicitly defined as the proportion of occupancy within the spatial grid of the sampling quadrat and its respective smaller units.

To illustrate how species abundances are computed, see Fig. 7, where we have considered a sampling quadrat size of 'S = 3' to simplify the figure. For example, in the visual example, within the sampling quadrat C_2, the yellow diamond symbol is present in all sampling units, resulting in an abundance value of 9/9 = 1. In contrast, the red triangle has an abundance of 1/9, highlighting its much lower occupation density within the same quadrat. Occasionally, certain quadrats may not have species records due to uneven sampling efforts, as shown in the figure density from GBIF (Fig. 6). The sampling quadrat C_1 in the figure illustrates this issue. In such cases, we exclude these quadrats

and do not compute abundances to avoid unreliable abundance calculations. Because abundances cannot be calculated at specific quadrat sizes due to sampling limitations, we developed a multiresolution approach that adaptively increases the sampling area in data-sparse regions. This method systematically expands the quadrat size until sufficient occurrence data is obtained, thereby increasing sampling coverage while maintaining the highest possible spatial resolution for each location.

Automatic selection of the optimal sampling quadrat size at different resolutions

In GBIF, the density of observations and sampling effort varies greatly. For example, many more records of a species are available from well-financed, data-sharing countries, making it impossible to use a single spatial window size as a sampling unit to calculate global abundances. Instead, we adopted a multiscale/multiresolution modeling strategy. These approaches were developed and successfully applied to address problems with significant features at multiple temporal and spatial scales, such as modeling geophysics, fluids, and solids[52].

We start formulating the quadrat sampling size with a side length based on the power of 2 of S, which is smaller, and with a fixed size of 5 × 5 km. The side length of each sampling quadrat is defined as $S = ISQS \cdot 2^n$, where $n \in \mathbb{Z}, 0 \le n \le 6$, and $ISQS$ is the initial sampling quadrat size. This formulation ensures that each sampling quadrat has a proportional side size and that all sizes of sampling units are univocally self-contained within each other (see Fig. 7).

The sample coverage estimation allows us to analytically describe the potential for a given species to be present within the sample, even if it was not detected during the sampling process. Recent studies using crowd-sourced data have shown that a relatively small number of samples is sufficient to achieve high levels of sampling completeness when diversity estimation focuses on dominant species[53]. We estimated the inventory completeness using the Chao and Jost sample coverage estimator[54] to target dominant species. This definition provides a bias-reduced sample completeness estimator:

$$C = 1 - \frac{f_1}{l}\left[\frac{(l-1)f_1}{(l-1)f_1 + 2f_2}\right] \tag{2}$$

where f_1 is the number of singletons (species represented by only one individual) in the sample, f_2 is the number of doubletons (species represented by only two individuals) in the sample l, the total number of individuals in the sample. The expression (2) quantifies the fraction of the total number of individuals in the assemblage that belong to the detected species, with each individual being proportionally weighted by its abundance[55]. We calculated the sample coverage for each sampling unit at all sizes to ensure that we captured the most abundant species. Finally, we considered the computed abundances provided by the smallest sampling unit, which had at least 80% completeness. This strategy aims to generate high-resolution abundance maps while addressing two critical challenges: optimizing spatial detail and mitigating sampling bias inherent in occurrence data. The method specifically ensures robust abundance estimates of dominant species in a given sampling quadrat size, even in regions where the sampling effort is heterogeneous. Fig. 7 illustrates how this process was carried out iteratively.

Iterative computation of trait distribution parameters from the species abundance maps

In addition to the mean, central moments, such as the variance, skewness, and kurtosis, are commonly used in descriptive statistics. In this study, we adopted the approach proposed by ref. 56 to derive generalized one-pass formulas that accommodate arbitrary orders and variables. It has been demonstrated that these formulas exhibit numerical stability and can be easily parallelized for enhanced scalability. Moreover, the researchers extended their findings to compute weighted or compound moments, a method we have employed in our analysis.

We follow a two-step process to compute the CWMs. Firstly, we calculate the mean and central moments of each PFT separately and iteratively, using the computed species abundances as weights. Next, we combined the moments calculated for each PFT using the same approach to compute the final moments, but using the fractional land cover map based on remote sensing as weights to provide final trait estimates.

We will follow the notation provided in ref. 56 where the weighted arithmetic mean ($\widetilde{x}_W$) with respect to a given set of weights (W) is computed using the following expressions:

$$W \triangleq \sum_{i=1}^{n} w_i,\ W_A \triangleq \sum_{i=1}^{n_A} w_i,\ W_B \triangleq \sum_{i=n_A+1}^{n} w_i, \tag{3}$$

where A and B are two multisets of respective sizes n_A and n_B, included in a finite population of n equiprobable values (x_i) in a multiset $S = \{x_i\}_{i=1}^{n}$. For example, in our case, A and B would be normalized abundances of two species computed from GBIF for a given PFT.

The weighted version of the mean and the pth ($p \in \mathbb{N}, p \ge 2$) central moment ($M_p$) of a random variable can be expressed as:

$$\widetilde{x}_W \triangleq \frac{1}{W}\sum_{i=1}^{n} w_i x_i, \tag{4}$$

$$M_p \triangleq \sum_{i=1}^{n} w_i (x_i - \widetilde{x}_W)^p, \tag{5}$$

which is expressed in their iterative form:

$$\widetilde{x}_W = \widetilde{x}_{W,A} + \frac{W_B}{W}\delta_{B,A}, \tag{6}$$

$$\begin{aligned} M_p = M_p^A + M_p^B + W_A\left(\frac{-W_B}{W}\delta_{B,A}\right)^p + W_B\left(\frac{W_A}{W}\delta_{B,A}\right)^p \\ + \sum_{k=1}^{p-2}\binom{p}{k}\delta_{B,A}^k\left[M_{p-k}^A\left(\frac{-W_B}{W}\right)^k + M_{p-k}^B\left(\frac{W_A}{W}\right)^k\right], \end{aligned} \tag{7}$$

where $\delta_{B,A} \triangleq \widetilde{x}_{W,B} - \widetilde{x}_{W,A}$.

Finally, to estimate the standard deviation(σ), skewness(SK), and excess of kurtosis(eK) from the central moments we use the following expressions:

$$\sigma = \sqrt{M_2},\ SK = \frac{M_3}{\sigma^3},\ eK = \frac{M_4}{\sigma^4} - 3. \tag{8}$$

The proposed approach has almost no fitted parameters; instead, it relies on species-level trait means fixed from TRY, and calculating pixel-level CWMs/TWMs is done through deterministic aggregation of empirically estimated abundances with remotely sensed PFT fractions. Consequently, excluding any spatial block (e.g., a continent) results in identical estimates, and standard spatial cross-validation approaches are not informative for this workflow.

Linking species abundance to expected mean traits. All considered plant measurements from the TRY database were transformed (natural logarithm) to normalize the distribution of data[18]. Accordingly, dispersion is quantified on the log scale: std[ln(trait)] is the geometric standard deviation and should be interpreted multiplicatively, i.e., variability on the original scale corresponds to factors of exp(±std) around the geometric mean. This convention also aligns with the

ln-transformed community means provided by sPlotOpen, ensuring that predictions and reference data are expressed on consistent units. The trait value t_i for a given species name was computed as the arithmetic mean of all TRY available transformed observations (TRY v5). Finally, due to the similar species name standardization process followed by GBIF and TRY[19,28], we linked the derived GBIF abundances via their taxonomic nomenclature. Moreover, we employed a fuzzy matching method based on Levenshtein distance metrics[57] to maximize the connections between the two databases. This association between databases allows access to existing functional trait information from the TRY database and, subsequently, enables the use of the extensive geospatial information from GBIF to better depict global trait distributions.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The plant functional trait maps and higher-order statistical moments generated in this study have been deposited in the Zenodo repository under accession code 18832368. An interactive visualization is available via Google Earth Engine (https://ee-almoma153.projects.earthengine.app/view/gbiftraitdescription) and through the Google Earth Engine Community Catalog (https://gee-community-catalog.org/). The TRY and sPlotOpen databases used in this study are openly accessible at their respective repositories: https://www.try-db.org, and https://www.idiv.de/splot. The biodiversity occurrence data used in this study were retrieved from GBIF.org; the data are available via the dataset citation provided in the References[58], https://doi.org/10.15468/dl.an9g68, https://doi.org/10.15468/dl.q65cat, and https://doi.org/10.15468/dl.75a8qk.

References

1. Reich, P. B., Wright, I. J. & Lusk, C. H. Predicting leaf physiology from simple plant and climate attributes: a global glopnet analysis. *Ecol. Appl.* **17**, 1982–1988 (2007).
2. Kattge, J. et al. Try–a global database of plant traits. *Glob. change Biol.* **17**, 2905–2935 (2011).
3. Homolova, L., Malenovsky`, Z., Clevers, J. G., García-Santos, G. & Schaepman, M. E. Review of optical-based remote sensing for plant trait mapping. *Ecol. Complex.* **15**, 1–16 (2013).
4. He, N. et al. Ecosystem traits linking functional traits to macroecology. *Trends Ecol. Evol.* **34**, 200–210 (2019).
5. Lavorel, S. & Grigulis, K. How fundamental plant functional trait relationships scale-up to trade-offs and synergies in ecosystem services. *J. Ecol.* **100**, 128–140 (2012).
6. Pappas, C., Fatichi, S. & Burlando, P. Modeling terrestrial carbon and water dynamics across climatic gradients: does plant trait diversity matter? *New. Phytol.* **209**, 137–151 (2016).
7. Wullschleger, S. D. et al. Plant functional types in earth system models: past experiences and future directions for application of dynamic vegetation models in high-latitude ecosystems. *Ann. Bot.* **114**, 1–16 (2014).
8. Anderegg, L. D. et al. Representing plant diversity in land models: an evolutionary approach to make "functional types" more functional. *Glob. Change Biol.* **28**, 2541–2554 (2022).
9. Mahecha, M. D. et al. Biodiversity and climate extremes: known interactions and research gaps. *Earth's. Future* **12**, e2023EF003963 (2024).
10. Reichstein, M., Bahn, M., Mahecha, M. D., Kattge, J. & Baldocchi, D. D. Linking plant and ecosystem functional biogeography. *Proc. Natl. Acad. Sci.* **111**, 13697–13702 (2014).
11. Anderegg, W. R., Trugman, A. T., Bowling, D. R., Salvucci, G. & Tuttle, S. E. Plant functional traits and climate influence drought intensification and land–atmosphere feedbacks. *Proc. Natl. Acad. Sci.* **116**, 14071–14076 (2019).
12. Butler, E. E. et al. Increasing functional diversity in a global land surface model illustrates uncertainties related to parameter simplification. *J. Geophys. Res.: Biogeosci.* **127**, e2021JG006606 (2022).
13. Scheiter, S., Langan, L. & Higgins, S. I. Next-generation dynamic global vegetation models: learning from community ecology. *New. Phytol.* **198**, 957–969 (2013).
14. Sakschewski, B. et al. Leaf and stem economics spectra drive diversity of functional plant traits in a dynamic global vegetation model. *Glob. Change Biol.* **21**, 2711–2725 (2015).
15. Enquist, B. J. et al. Scaling from traits to ecosystems: developing a general trait driver theory via integrating trait-based and metabolic scaling theories. *Adv. Ecol. Res.* **52**, 249–318 (2015).
16. Huang, C., Xu, Y. & Zang, R. Variation patterns of functional trait moments along geographical gradients and their environmental determinants in the subtropical evergreen broadleaved forests. *Front. Plant Sci.* **12**, 686965 (2021).
17. Fisher, R. A. et al. Vegetation demographics in earth system models: a review of progress and priorities. *Glob. change Biol.* **24**, 35–54 (2018).
18. Kattge, J. et al. Try plant trait database–enhanced coverage and open access. *Glob. change Biol.* **26**, 119–188 (2020).
19. Sabatini, F. M. et al. splotopen–an environmentally balanced, open-access, global dataset of vegetation plots. *Glob. Ecol. Biogeogr.* **30**, 1740–1764 (2021).
20. Dechant, B. et al. Intercomparison of global foliar trait maps reveals fundamental differences and limitations of upscaling approaches. *Remote Sens. Environ.* **311**, 114276 (2024).
21. van Bodegom, P. M., Douma, J. C. & Verheijen, L. M. A fully traits-based approach to modeling global vegetation distribution. *Proc. Natl. Acad. Sci.* **111**, 13733–13738 (2014).
22. Butler, E. E. et al. Mapping local and global variability in plant trait distributions. *Proc. Natl. Acad. Sci. USA* **114**, E10937–E10946 (2017).
23. Madani, N. et al. Future global productivity will be affected by plant trait response to climate. *Sci. Rep.* **8**, 2870 (2018).
24. Moreno-Martínez, Á et al. A methodology to derive global maps of leaf traits using remote sensing and climate data. *Remote Sens. Environ.* **218**, 69–88 (2018).
25. Boonman, C. C. et al. Assessing the reliability of predicted plant trait distributions at the global scale. *Glob. Ecol. Biogeogr.* **29**, 1034–1051 (2020).
26. Schiller, C., Schmidtlein, S., Boonman, C., Moreno-Martínez, Á & Kattenborn, T. Deep learning and citizen science enable automated plant trait predictions from photographs. *Sci. Rep.* **11**, 16395 (2021).
27. Vallicrosa, H. et al. Global maps and factors driving forest foliar elemental composition: the importance of evolutionary history. *New Phytol.* **233**, 169–181 (2022).
28. Wolf, S. et al. Citizen science plant observations encode global trait patterns. *Nat. Ecol. Evol.* **6**, 1125–1134 (2022).
29. Potts, J. M. & Elith, J. Comparing species abundance models. *Ecol. Model.* **199**, 153–163 (2006).
30. Waldock, C. et al. A quantitative review of abundance-based species distribution models. *Ecography* **2022**, e05694 (2022).
31. Dickinson, J. L., Zuckerberg, B. & Bonter, D. N. Citizen science as an ecological research tool: challenges and benefits. *Annu. Rev. Ecol. Evol. Syst.* **41**, 149–172 (2010).
32. Wright, I. J. et al. The worldwide leaf economics spectrum. *Nature* **428**, 821–827 (2004).
33. Díaz, S. et al. The global spectrum of plant form and function. *Nature* **529**, 167–171 (2016).
34. Jetz, W., McPherson, J. M. & Guralnick, R. P. Integrating biodiversity distribution knowledge: toward a global map of life. *Trends Ecol. Evol.* **27**, 151–159 (2012).

35. Shiklomanov, A. N. et al. Does the leaf economic spectrum hold within plant functional types? A Bayesian multivariate trait meta-analysis. *Ecol. Appl.* **30**, e02064 (2020).
36. Fotheringham, A. S. & Wong, D. W. The modifiable areal unit problem in multivariate statistical analysis. *Environ. Plan. A* **23**, 1025–1044 (1991).
37. Fithian, W., Elith, J., Hastie, T. & Keith, D. A. Bias correction in species distribution models: pooling survey and collection data for multiple species. *Methods Ecol. Evol.* **6**, 424–438 (2015).
38. Le Bagousse-Pinguet, Y. et al. Functional rarity and evenness are key facets of biodiversity to boost multifunctionality. *Proc. Natl. Acad. Sci.* **118**, e2019355118 (2021).
39. Zurita-Milla, R., Laurent, V. & van Gijsel, J. Visualizing the ill-posedness of the inversion of a canopy radiative transfer model: a case study for sentinel-2. *Int. J. Appl. Earth Obs. Geoinf.* **43**, 7–18 (2015).
40. Bruelheide, H., Dengler, J., Purschke, O. & Lenoir, J. Global trait-environment relationships of plant communities. *Nat. Ecol. Evol.* **2**, 1906–1917 (2018).
41. Zhang, X. et al. Glc_fcs30d: the first global 30-m land-cover dynamic monitoring product with a fine classification system from 1985 to 2022 using dense time-series landsat imagery and continuous change-detection method. *Earth Syst. Sci. Data Discuss.* **2023**, 1–32 (2023).
42. Zanaga, D. et al. ESA WorldCover 10 m 2021 v200. *Zenodo* https://doi.org/10.5281/zenodo.7254221 (2022).
43. Grime, J. Benefits of plant diversity to ecosystems: immediate, filter and founder effects. *J. Ecol.* **86**, 902–910 (1998).
44. Maestre, F. T. et al. Grazing and ecosystem service delivery in global drylands. *Science* **378**, 915–920 (2022).
45. Beck, J., Böller, M., Erhardt, A. & Schwanghart, W. Spatial bias in the gbif database and its effect on modeling species' geographic distributions. *Ecol. Inform.* **19**, 10–15 (2014).
46. Hwang, W.-H. & He, F. Estimating abundance from presence/absence maps. *Methods Ecol. Evol.* **2**, 550–559 (2011).
47. Joseph, L. N., Field, S. A., Wilcox, C. & Possingham, H. P. Presence–absence versus abundance data for monitoring threatened species. *Conserv. Biol.* **20**, 1679–1687 (2006).
48. Barry, S. C. & Welsh, A. H. Generalized additive modelling and zero inflated count data. *Ecol. Model.* **157**, 179–188 (2002).
49. Guisan, A., Theurillat, J.-P. & Kienast, F. Predicting the potential distribution of plant species in an alpine environment. *J. Veg. Sci.* **9**, 65–74 (1998).
50. Zizka, A. et al. Coordinatecleaner: standardized cleaning of occurrence records from biological collection databases. *Methods Ecol. Evol.* **10**, 744–751 (2019).
51. de Oliveira, A. C. P. et al. From species presences to abundances: using unevenly collected plant species presences to disclose the structure and functioning of a dryland ecosystem. *Ecol. Indic.* **113**, 106221 (2020).
52. Weinan, E. *Principles of Multiscale Modeling*. Cambridge University Press (2011).
53. Callaghan, C. T. et al. Quantifying effort needed to estimate species diversity from citizen science data. *Ecosphere* **13**, e3966 (2022).
54. Chao, A. & Jost, L. Coverage-based rarefaction and extrapolation: standardizing samples by completeness rather than size. *Ecology* **93**, 2533–2547 (2012).
55. Chao, A. et al. Quantifying sample completeness and comparing diversities among assemblages. *Ecol. Res.* **35**, 292–314 (2020).
56. Pébay, P., Terriberry, T. B., Kolla, H. & Bennett, J. Numerically stable, scalable formulas for parallel and online computation of higher-order multivariate central moments with arbitrary weights. *Comput. Stat.* **31**, 1305–1325 (2016).
57. Levenshtein, V. I. Binary codes capable of correcting deletions, insertions, and reversals. *Sov. Phys. Dokl.* **10**, 707–710 (1966).
58. GBIF.org. *GBIF Occurrence Download* https://doi.org/10.15468/dl.3z8x8v (2022).
59. Fisher, R. A. *Statistical Methods for Research Workers* (Oliver and Boyd, Edinburgh, 1932).

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Author contributions

Á.M.M. conceived the study, designed the methodology, and led the writing of the manuscript. Á.M.M., J.M.M., and J.E.A. performed the formal analyses, with J.M.M. and J.E.A. also contributing to the study design. F.M.S. and J.K. provided expert insights regarding the sPlotOpen and TRY datasets, respectively. G.C.V., B.D., F.D.S., E.E.B., P.M.V., T.K., J.K., and M.D.M. made substantial contributions to drafting the manuscript. Finally, J.M.M, J.E.A., B.D., J.K., F.D.S., G.D., P.M.V., E.E.B., M.D.M., E.I.V., T.K., J.P., G.B., R.P.P., P.A.T., N.C., D.L., and G.C.V. contributed conceptual ideas and critically reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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